WORKSHEET 3.1 : VECTORS & PARAMETRIC EQUATION OF LINES IN $\mathbb{R}^2$

1) direction vector $\vec{m}$
parameter t

Equation of a line

$$\vec{r} = \vec{r_0} + t\vec{m}$$

$\vec{r_0}$: starting location; t: # of steps; $\vec{m}$: direction

2a) $\vec{m} = (-3, 1)$

b) $\vec{m} = \vec{AB}$
$= (-8, -10)$ or $(-4, -5)$ or $(4, 5)$

c) $m = \frac{3}{1}$ (slope)
$\vec{m} = (1, 3)$

d) $\vec{m} = (4, 3)$

e) $\vec{m} = (1, 0)$ parallel to x-axis

f) $\vec{m} = (0, 1)$ parallel to y-axis

any scalar multiples.

3a) $x = 3 - 8t$
$y = 4t$

if $t = 0$: $(3, 0)$

if $t = 1$: $(-5, 4)$

b) $\vec{r} = (4, 0) + t(0, 5)$

if $t = 0$: $(4, 0)$

if $t = 1$: $(4, 5)$

see reasons.

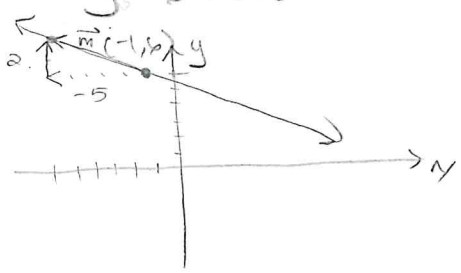
	y-int	m = slope
4a) $\vec{r} = (0,-1) + t(2,1)$, $t \in \mathbb{R}$	$(0,-1)$	$m = \frac{1}{2}$
b) $\vec{r} = (0,3) + t(1,-1)$, $t \in \mathbb{R}$	$(0,3)$	$m = \frac{-1}{1}$

5a) $x = -1 - 5t$

$y = 6 + 2t$

$\vec{m} = (-5, 2)$

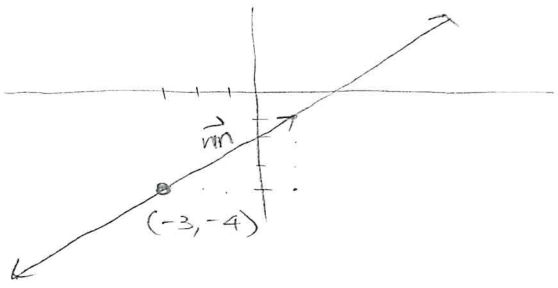
pt $(-1, 6)$



b) $\vec{r} = (-3,-4) + t(4,3)$

$\vec{m} = (4,3)$

pt $(-3,-4)$



6a) $x = 1 + 4t$, $t \in \mathbb{R}$

$y = 1 - 4t$

Let $t = 1$ $(x,y) = (5,-3)$

Let $t = 2$ $(x,y) = (9,-7)$

b) $x = 5 + t$, $t \in \mathbb{R}$

$y = 3t$

Let $t = 1$ $(x,y) = (6,3)$

Let $t = 2$ $(x,y) = (7,6)$

7a) point $(0,0)$

$\vec{m} = (1,0)$

$x = t$

$y = 0$, $t \in \mathbb{R}$

b) point $(0,1)$ ← any point not on x-axis

$\vec{m} = (1,0)$

$x = t$

$y = 1$, $t \in \mathbb{R}$

8a) $\vec{r} = (-2,7) + t(3,-4)$, $t \in \mathbb{R}$

b) $\vec{r} = \left(2, \frac{3}{4}\right) + t\left(\frac{2}{3}, 6\right)$, $t \in \mathbb{R}$

or $\vec{m} = (2,18)$ $\Rightarrow$ $\vec{r} = \left(2, \frac{3}{4}\right) + s(2,18)$, $s \in \mathbb{R}$

c) $\vec{r} = (1,-1) + t(-\sqrt{3}, 3)$, $t \in \mathbb{R}$

d) $\vec{r} = (0,0) + t(-2,3)$, $t \in \mathbb{R}$

$\vec{r} = t(-2,3)$, $t \in \mathbb{R}$

10 a) $\vec{m_1} = (-3,4)$ $\vec{m_2} = (-4,-3)$

$\vec{m_1} \cdot \vec{m_2} = 12 - 12 = 0$ $\therefore$ $\perp$ lines

b) $\vec{m_1} = (-3,4)$ $\vec{m_2} = (3,-4)$

$\vec{m_1} = -\vec{m_2}$ $\therefore$ collinear / parallel

c) $\vec{m_1} = (-3,4)$ $\vec{m_2} = (4,-3)$

check: $\vec{m_1} = k\vec{m_2}$

$(-3,4) = k(4,-3)$

$-3 = 4k$ $\quad 4 = -3k$

$\frac{-3}{4} = k$ $\quad \frac{-4}{3} = k$

not the same.

$\therefore$ not collinear

$\vec{m_1} \cdot \vec{m_2} = -12 - 12$

$= -24 \neq 0$.

$\therefore$ not perpendicular

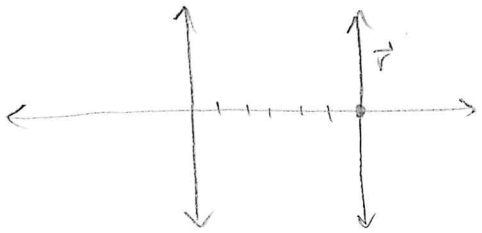
11) $\vec{r_0} = (4,5)$

$\perp$ to $(3,7)$

$\therefore \vec{m} = (7,-3)$

$\therefore \vec{r} = (4,5) + t(7,-3)$, $t \in \mathbb{R}$

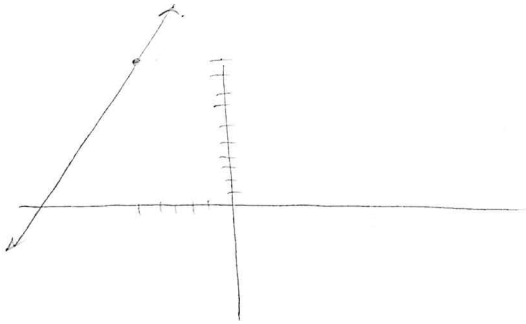
12 a) $x = 6 \quad y = 1 + 7t$



x-int = 6.

y-int none

b) $\vec{r} = (-5, 10) + t(1, 5)$



using parametric equations

x-int

$(x, 0)$

$(x, 0) = (-5, 10) + t(1, 5)$

$x = -5 + t \qquad 0 = 10 + 5t$

$x = -5 - 2 \leftarrow -2 = t$

$\boxed{x = -7}$

y-int

$(0, y)$

$(0, y) = (-5, 10) + t(1, 5)$

$0 = -5 + t \rightarrow y = 10 + 5t$

$t = 5 \rightarrow y = 10 + 5(5$

$\boxed{y = 35}$

OR //

$y = mx + b.$

$10 = 5(-5) + b$

$10 = -25 + b$

$35 = b$

$\therefore y = 5x + 35$

$\hookrightarrow \boxed{y\text{-int} = 35}$

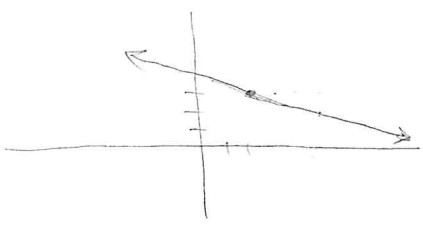
x-int => let $y = 0$

$0 = 5x + 35$

$-35 = 5x$

$\boxed{-7 = x}$

c) $\vec{r} = (2, 3) + t(3, -1)$



x-int

$(x, 0)$

$(x, 0) = (2, 3) + t(3, -1)$

$x = 2 + 3t \qquad 0 = 3 - t$

$x = 2 + 3(3) \leftarrow t = 3$

$\boxed{x = 11}$

y-int

$(y, 0)$

$(0, y) = (2, 3) + t(3, -1$

$0 = 2 + 3t \rightarrow y = 3 - t$

$\frac{-2}{3} = t \rightarrow y = 3 - (\frac{-2}{3}$

$\boxed{y = \frac{11}{3}}$

13) If $(1,4)$ is a point on the line, then there exists an unique t & u for each equation.

$(1,4) = (3,7) + t(2,5)$

$1 = 3 + 2t$ $\quad 4 = 7 + 5t$

$-2 = 2t$ $\quad -5 = 5t$

$-1 = t$ $\quad -1 = t$

SAME

$\therefore \vec{r_1}$ contains $(1,4)$

$(1,4) = (-5,6) + u(3,-1)$

$1 = -5 + 3u$ $\quad 4 = 6 - u$

$6 = 3u$ $\quad -2 = -u$

$2 = u$ $\quad 2 = u$

same

$\vec{r_2}$ contains $(1,4)$

$\vec{m_1} = (2,5)$

$\vec{m_2} = (3,-1)$

$$\cos\theta = \frac{\vec{m_1} \cdot \vec{m_2}}{|\vec{m_1}||\vec{m_2}|}$$

$$\cos\theta = \frac{6-5}{\sqrt{29}\sqrt{10}}$$

$$\cos\theta = \frac{-1}{\sqrt{290}}$$

$$\boxed{\theta = 87°}$$

17a) $F = (7,3) + t(2,-5)$, $t \in \mathbb{R}$

Check $P(5,8)$

$(5,8) = (7,3) + t(2,-5)$

$5 = 7 + 2t$ $\quad 8 = 3 - 5t$

$-2 = 2t$ $\quad 5 = -5t$

$-1 = t$ $\quad -1 = t$

SAME

$\therefore$ P is on the line

check $Q(17,-22)$

$(17,-22) = (7,3) + t(2,-5)$

$17 = 7 + 2t$ $\quad -22 = 3 - 5t$

$10 = 2t$ $\quad -25 = -5t$

$5 = t$ $\quad 5 = t$

SAME

$\therefore$ Q is on the line

b) since when $t=-1 \Rightarrow P(5,8)$

when $t=5 \Rightarrow Q(17,-22)$

$\therefore \vec{r} = (7,3) + t(2,-5), \quad -1 \leq t \leq 5$

OR $\vec{r} = (5,8) + t(2,-5), \quad 0 \leq t \leq 6.$

start

change the parameter (still 6 "steps")